

Student Enrollment & Course Management System (SECMS)

Phase 3 – Project Design
Built on the Salesforce Platform

Project	Details
Project Name	Student Enrollment & Course Management System (SECMS)
Platform	Salesforce (Lightning Experience, Developer Edition)
Team Member	Member 1: Vasanthakumar R Team Lead Member 2: Saravanan V Member 3: Syed Kutbudeen S Member 4: Raji P Member 5: Sriram S
Phase	3 of 8 – Project Design

1. Design Overview

SECMS is designed as a declarative Salesforce application. Data is stored in five custom objects, connected through lookup and master-detail relationships. Business rules are enforced with validation rules, an approval process and Flow, and users interact through a Lightning app with tabs, page layouts, reports and a dashboard.

2. System Architecture

Layer	Components
Presentation Layer	Lightning App (SECMS), Tabs, Page Layouts, Reports, Dashboard
Business Logic Layer	Validation Rules, Flow Automation, Approval Process
Data Layer	Custom Objects: Student, Course, Instructor, Enrollment, Payment
Security Layer	Profiles, Roles, Permission Sets, Field-level Security
Platform	Salesforce Cloud (multi-tenant, browser access)
Insert System-Architecture.png here	

3. Data Model – Objects and Fields

Object	Field	Type
Student	Student Name	Text / Auto Number
	Email	Email
	Phone	Phone
	Department	Picklist
	Status	Picklist
Course	Course Name	Text
	Course Code	Text (Unique)
	Fee	Currency
	Seats	Number
	Duration	Text / Number
Instructor	Instructor	Lookup (Instructor)
	Instructor Name	Text
	Email	Email
Enrollment	Department	Picklist
	Student	Master-Detail / Lookup (Student)
	Course	Lookup (Course)

	Enrollment Date	Date
	Status	Picklist
Payment	Enrollment	Lookup (Enrollment)
	Amount	Currency
	Payment Date	Date
	Payment Status	Picklist

4. Relationships

Parent	Child	Type	Meaning
Student	Enrollment	One-to-Many	One student can have many enrollments
Course	Enrollment	One-to-Many	One course can have many enrollments
Instructor	Course	One-to-Many	One instructor can teach many courses
Enrollment	Payment	One-to-Many	One enrollment can have payment records

Enrollment is the junction object between Student and Course (many-to-many).

5. Validation Rule Design

Object	Rule	Purpose
Student	Email and phone must be in valid format	Avoid wrong contact data
Course	Fee cannot be negative; seats must be greater than 0	Keep course data valid
Enrollment	Enrollment date cannot be in the future	Keep dates accurate
Payment	Amount must be greater than 0	Avoid invalid payments

6. Automation Design

- **Approval Process:** New enrollment is submitted → Administrator approves or rejects → status is updated.
- **Flow:** Record-triggered flow updates enrollment / payment status and sends notification on change.

7. User Interface Design

- **Lightning App:** SECMS app with navigation tabs for Students, Courses, Instructors, Enrollments, Payments, Reports and Dashboards.

- **Page Layouts:** Key fields at the top and related lists (Enrollments, Payments) below.
- **Reports:** Enrollments by course, student list by department, payment status.
- **Dashboard:** Charts for total enrollments, course-wise students and paid vs pending payments.

8. Security Design

Role / Profile	Access
System Administrator	Full access to all objects, flows and approvals
Admin / Office Staff	Create, read and edit on all SECMS objects
Instructor	Read-only on Course and Enrollment
Management / HOD	Read-only; access to reports and dashboards